Modifying LeanCoP Options On the Fly

(aka "the PAAR paper")

Introduction and Research

LeanCoP inspired a large family of connection provers that improve performance and completeness by making a small change to the control mechanism [6].

LeanCoP 2.0 [7] added 3 control options to the original LeanCoP: `cut` and `scut`, to restrict backtracking in connection calculus, and `comp(J)`, which makes iterative deepening complete by restarting the search and increasing the maximum height of this tree 1 at a time until it reaches maximum height `J`, and restarting from height 1 and with cuts disabled after exhausting the final tree.

This work gathers ideas from other implementations of connection provers from different domains.

Letz and Stenz [4] show that, in tableau search, it's preferable to have subgoals with fewer solutions to detect dead ends earlier. Reordering clauses to prioritise shorter clauses is a valid principle, because the number of solutions is roughly the product over the clause's subgoals.

Zombori, Urban, and Olšák [9] show that the entropy of a learned policy's action distribution is a key factor in guiding connection provers: policies whose entropy is too low tend to overcommit to narrow search paths, while policies whose entropy is too high spread effort too thinly. I adapt this insight at runtime by measuring the Shannon entropy of heuristic action scores at each proof step and use low entropy as a signal to add a `cut` to the current connection action.

Li et al. [5] frame the reset choice in SAT solvers as a multi-armed bandit problem. They show that recent search productivity should influence reset behaviour. Similar restart strategies for SAT solvers are also surveyet in Holden [2].

Novel Idea

Triggering the three control atoms in LeanCoP, `cut`, `scut`, and `comp(I)`, dynamically during the tree search rather than at the beginning can allow a prover to solve a tableau tree in less time and fewer steps.

A possible solution would be to use reinforcement learning, like RLCop [3] or learnCoP [citation coming soon!]. However, inference in these systems is slow and a heuristic-based solution might solve problems more quickly on a subset of TPTP problems.

Implementation

I use the existing connection prover implementation as a base for this code [8]. This allows me to use the existing code for hard things like Prolog parsing, and also to use PyCoP as a control group.

My code implements 6 different strategies.

FirstAction *(control)*
: Run the first legal action (PyCoP default).

LeanCoPCut *(control)*
: Run `cut`, `scut`, and `comp(7)`, like LeanCoP 2.0.

HeuristicSelection
: Choose reductions if possible, otherwise choose the extension with the shortest clause.

EntropyCut
: Run the first legal action, but add a cut when heuristic action scores have low entropy.

AdaptiveReset
: Resets the tree if it hits 5 dead-ends in a row.

SmartCut
: Combines the last three strategies.

Preliminary Results

A subset of 247 "representative" TPTP problems was used for evaluation [1]. The results with any significance are found in Table 1.

The comparer runs all 6 policies in parallel, with a 30-second time limit and a 50 000 step limit. A larger experiment, with the entirety of TPTP and a longer time limit, will come in the future.

The results show that all three policies have marked improvements in certain problems. It's likely that with further exploration we might get even better results than the control groups.

Problem	FirstAction	LeanCopCut	HeuristicSelection	EntropyCut	AdaptiveReset	SmartCut
GEO179+3	11 987	4744	9629	11 979	2546	max-steps
GEO181+3	11 929	4778	9812	11 925	max-steps	timeout
HWV026-3	10 174	4212	max-steps	10 174	10 174	max-steps
KRS095+1	552	504	952	548	552	1014
LCL186-1	245	477	297	245	368	timeout
LCL644+1.001	8	8	2	8	8	2
LCL679+1.001	1	1	1	1	1	1
MSC015-1.005	25 401	29 037	26 257	28 251	7556	8816
NUM291+1	32 860	19 936	max-steps	32 860	32 860	max-steps
NUM300+1	11 135	7471	18 308	11 135	11 135	18 306
NUM615+1	max-steps	42 778	max-steps	max-steps	max-steps	max-steps
NUM634+3	timeout	timeout	2	timeout	timeout	2
SET062+3	24	24	26	24	24	26
SET175+3	max-steps	12 401	max-steps	max-steps	1062	1031
SET603+3	max-steps	9953	max-steps	max-steps	timeout	853
SET626+3	max-steps	35 231	max-steps	max-steps	timeout	timeout
SET886+1	max-steps	24 582	max-steps	max-steps	timeout	timeout
SET904+1	105	101	140	101	105	140
SEU121+2	max-steps	34 536	max-steps	max-steps	timeout	timeout
SWV232+1	54	54	timeout	54	54	timeout
SWV248-2	timeout	timeout	7016	6018	timeout	timeout
SWV264-2	9	9	11	9	9	11
SYN061-1	46	46	46	46	46	46
SYN582-1	15 843	11 650	40 624	15 430	timeout	timeout
SYN663-1	5651	max-steps	5588	5626	timeout	max-steps
SYN709-1	16 642	5238	11 201	16 065	max-steps	max-steps
SYN726-1	max-steps	max-steps	4468	max-steps	timeout	max-steps

Table 1: Amount of steps required to solve representative TPTP problems. Green numbers show best results that are significantly better than the control groups. Only 27 of these problems finished within the time and step limit; larger experiments with a higher time limit will give us more information.

References

[1] Karel Chvalovský and Jan Jakubův. "Characteristic Subsets of TPTP Benchmarks". In: *Proceedings of the 6th Conference on Artificial Intelligence and Theorem Proving (AITP 2021)* (Aussois, France, Sept. 5–10, 2021). Extended abstract in informal proceedings. 2021.

[2] Sean B. Holden. "Machine Learning for Automated Theorem Proving: Learning to Solve SAT and QSAT". In: *Foundations and Trends® in Machine Learning* 14.6 (2021), pp. 807–989. ISSN: 1935-8237, 1935-8245.

[3] Cezary Kaliszyk et al. "Reinforcement Learning of Theorem Proving". In: *Advances in Neural Information Processing Systems*. Ed. by S. Bengio et al. Vol. 31. Curran Associates, Inc., 2018, pp. 8836–8847.

[4] Reinhold Letz and Gernot Stenz. "Model Elimination and Connection Tableau Procedures". In: *Handbook of Automated Reasoning*. Elsevier, 2001, pp. 2015–2114. ISBN: 978-0-444-50813-3.

[5] Chunxiao Li et al. "A Reinforcement Learning Based Reset Policy for CDCL SAT Solvers". In: *CoRR* abs/2404.03753 (Apr. 19, 2024). arXiv: 2404.03753 [cs.LO]. Pre-published.

[6] Jens Otten. "20 Years of leanCoP — an Overview of the Provers". In: *Proceedings of the 1st International Workshop on Automated Reasoning with Connection Calculi (AReCCa@TABLEAUX 2023)*. Vol. 3613. CEUR Workshop Proceedings. 2023.

[7] Jens Otten. "Restricting Backtracking in Connection Calculi". In: *AI Communications* 23.2–3 (2010), pp. 159–182. ISSN: 09217126.

[8] Fredrik Rømming, Jens Otten, and Sean B. Holden. "Connections: Markov Decision Processes for Classical, Intuitionistic and Modal Connection Calculi". In: *Proceedings of the 1st International Workshop on Automated Reasoning with Connection Calculi*. Vol. 3613. CEUR Workshop Proceedings. 2023, pp. 107–118.

[9] Zsolt Zombori, Josef Urban, and Miroslav Olšák. "The Role of Entropy in Guiding a Connection Prover". In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Ed. by Anupam Das and Sara Negri. Vol. 12842. Cham: Springer International Publishing, 2021, pp. 218–235. ISBN: 978-3-030-86058-5 978-3-030-86059-2.